

1. How does the KNN algorithm work?

☒ It finds the K nearest data points (neighbors) to a query point and classifies it based on the majority class among them.

2. How do you choose the right K?

☒ Try different values using cross-validation; odd K is preferred for binary classification to avoid ties.

3. Why is normalization important in KNN?

☒ KNN uses distance; without normalization, larger-scale features dominate the result.

4. What is the time complexity of KNN?

☒ **Training:** $O(1)$ (lazy learner), **Prediction:** $O(n \times d)$ where n = training samples, d = features.

5. What are pros and cons of KNN?

☒ 1) Simple, no training phase, good with small datasets.
2) Slow on large data, sensitive to irrelevant/noisy features.

6. Is KNN sensitive to noise?

☒ Yes, noisy or irrelevant features can mislead distance calculations.

7. How does KNN handle multi-class problems?

☒ It counts the class frequency among neighbors and picks the class with the highest count.

8. What's the role of distance metrics in KNN?

☒ Distance metrics (e.g., Euclidean, Manhattan) determine which points are "nearest"; crucial for accuracy.